MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION November 2025

First SEMESTER

Skill Development Components (SDC) - MC1SDCREM102 - MATHEMATICS FOR RENEWABLE ENERGY- I

(2025 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes Maximum Marks: 50

Remember(*IN), tfnd'erstand{If}, *Apply(A), Analyse(An),* Eva/t/ate{E}, *Create(C),* Sk//{S}, *Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer

Answer any 15 questions

Each question carries 1 marks

1. Find the HCF of 8 and 12 **[K] / (CO1]**

4

2. Simplify: (5/6) divided by (10/9) **[K] / [CO1]**

3/4

3. What is the rank of a matrix? **[A] / |CO3]**

The rank is the maximum number of linearly independent rows (or columns) in the matrix. It indicates the dimension of the vector space spanned by its rows or columns.

4. What is the condition for the multiplication of two matrices to be possible? [A] / |CO3]

Multiplication is possible only if the number of **columns in the first matrix** is equal to the number of **rows in the second matrix**.

5. Give an Example for Lower Triangular Matrix of order 3x3 **[A] / |CO3]**

Any matrix where all numbers *above* the main diagonal are zero.

6. What is the determinant of matrix **[A] / |CO3]**

30

7. What is the order of a 2x3 matrix? **[A] / |CO3]**

It has 2 rows and 3 columns.

8. Define scalar multiplication of a matrix. **[An] / |CO3]**

Scalar multiplication means multiplying every single element inside a matrix by a specific real number (called a scalar).

K*(ANY MATRIX)

9. What is the Value Sin 60° **[An] / CO4]**

$= \frac{\sqrt{3}}{2}$ (approx. 0.866).

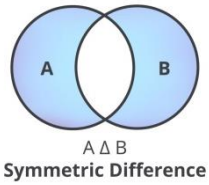
10. Define Tan theta in terms of Sin and Cos **[An] / [CO4]**

Tan(Φ) = Sin(Φ) / Cos(Φ).

11. A set with no elements in it is called? **[U] / [COS]**

Empty Set (also called Null Set or Void Set).

12. What is the symmetric difference of sets A and B **[U] / [COS]**



The set of elements that are in A or in B, but **not** in both.

A delta B = (A - B) U (B - A).

13. Identify if the set of positive integers is finite or infinite **[U] / [COS]**

Infinite.

14. State the idempotent law for intersection **[U] / [COS]**

The intersection of a set with itself is simply the set itself. (A ∩ A = A).

15. Find the power set of {1, 2}. **[U] / [COS]**

The set of all subsets: { {}, {1}, {2}, {1, 2} }.

16. What does "U" stand for in set theory? **[U] / [COS]**

"U" typically stands for the **Universal Set**.

17. If A - B = B - A then A and B are... **[U] / [COS]**

Equal sets (A = B). This is because the only way the difference of two sets can be identical is if both differences are empty.

Part B

Short Essay Answer any 3 questions

Each question carries 5 marks

18. Write any three properties of identity matrix. **[A] / [CO3]**

1. Multiplicative Identity: A * I = A (Multiplying by Identity leaves the matrix unchanged).
2. Inverse: The inverse of an identity matrix is the identity matrix itself.
3. Determinant: The determinant of an identity matrix is always 1.

19. Show that (A inverse) = I with an example. **[A] / [CO3]**

- **Any matrix multiplied their inverse should be I**

  The result is the Identity Matrix (I).

20. Find the period and frequency of y = cos(2x + 45°) **[An] / [CO4]**

Period = 360 / 2 = **180°**.

Frequency = **1 / 180**.

21. Derive the general equation of circle from its standard form. **[An] / [CO4]**

Standard form: $(x - h)^2 + (y - k)^2 = r^2$

General Equation: $(x)^2 + (y)^2 = r^2$

22. Explain the difference between roster form and tabular form of sets with examples.

[U) / (CO5]

1. A = {a, e, i, o, u}.

2. {x | x is a vowel}

Part C

Essay

Answer any 2 questions Each question carries 10 marks

23. Simplify the following expression step by step and find the final answer [K] / [CO1]

Answer = **14.33** (or 43/3).

24. For matrices A=[1 2; 3 4] and B=[2 0; 1 2], verify that (AB)^T = B^T A^T.

[A] / (CO3]

Step 1: Calculate Product AB = Matrix AB = $\begin{matrix} 4 & 4 \\ 10 & 8 \end{matrix}$

Step 2: Find Left Side (Transpose of AB)

(AB)^T = [$\begin{matrix} 4 & 10 \\ 4 & 8 \end{matrix}$]

Step 3: Find Right Side (B^T multiplied by A^T)

Transpose of B = $\begin{matrix} 2 & 1 \\ 0 & 2 \end{matrix}$ [

Transpose of A = $\begin{matrix} 1 & 3 \\ 2 & 4 \end{matrix}$

Multiply B^T * A^T:

B^T A^T = $\begin{matrix} 4 & 4 \\ 10 & 8 \end{matrix}$

Both sides are equal. Verified.

25. For the function y = 5 sin(2x + 45°), determine amplitude, frequency, phase shift, and Time Period. **[An] / [CO4]**

Amplitude:. **5** **Time Period:** 360 / 2 = **180°**

Frequency: 1/180 **Phase Shift:** Solve.**-22.5°**

26. State and prove distributive laws of union and intersection of sets with examples.

[U) / (CO5]

With any example matrix prove the Laws

Law 1: Union distributes over Intersection

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Law 2: Intersection distributes over Union

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$